Lecture 15 — Midterm 1 prep

Wed Sep 30

Last updated: October 8, 2026.

Textbook: Boyd & Vandenberghe, *VMLS* (PDF)

No separate lecture notes for today. Class time is Midterm 1 preparation using the posted study guide and practice packet. Related objectives: Lectures. Schedule.

Prepare / work in class

1. Midterm 1 study guide — mark each outcome confident / shaky / blank; build your definitions and formulas list.
2. Midterm 1 practice — definitions, computations, and harder mixed items (answers at the end).

Bring questions from Quizzes 1–4 and Lab 0. Midterm 1 is Friday, October 2 (entire class period; the exam itself is not posted online).

Learning objectives

By the end of class, you should be able to:

1. Connect distance, similarity, span, and independence.
2. Select an appropriate vector tool for a new problem.
3. Move between algebraic, geometric, computational, and contextual representations.
4. Diagnose and explain common errors.
5. Identify gaps and make a targeted study plan before Friday.